DATA WAREHOUSING AND MINING LABORATORY

Experiment No. 4: Apply Data Transformation and Normalization Techniques Using WEKA

Aim: To apply data transformation and normalization techniques on a dataset using WEKA and study their importance in data preprocessing for data mining applications.

Objectives: After successful completion of this experiment, students will be able to:

1. Understand the concept of data transformation.
2. Understand the need for data normalization.
3. Differentiate between data cleaning and data transformation.
4. Apply normalization techniques using WEKA.
5. Compare datasets before and after transformation.
6. Analyze the impact of normalization on attribute values.
7. Prepare datasets for classification and clustering algorithms.

Importance of this Experiment:

Data collected from various sources often contains attributes having different value ranges.

For example:

Attribute	Value
Age	5
Weight	15
Income	50000

In such cases, attributes with large values may dominate attributes with small values during analysis. Data transformation and normalization are therefore important preprocessing techniques that help bring all attributes to a comparable scale and improve the performance of data mining algorithms.

Expected Outcome: After completing this experiment, Students will be able to:

1. Transform numerical data into a suitable format for analysis.
2. Apply normalization techniques using WEKA.

3. Compare original and transformed datasets.
4. Understand the importance of attribute scaling.
5. Prepare datasets for classification and clustering experiments.

Facilities Required:

Hardware Requirements:

- Personal Computer / Laptop
- Intel i3 Processor or higher
- Minimum 4 GB RAM
- Internet Connection (Optional)

Software Requirements:

- Windows 10/11
- WEKA Tool
- Sample Dataset (student_transform.arff)

Prerequisite Knowledge: Students should be familiar with:

- Experiment No. 1: Installation and Exploration of WEKA Tool and Available Datasets
- Experiment No. 2: Create and Analyze ARFF Files Using WEKA
- Experiment No. 3: Perform Data Cleaning and Missing Value Treatment Using WEKA
- Data Cleaning Concepts
- Attributes and Instances
- Data Mining Fundamentals

Theory:

1. Data Transformation

Data Transformation is the process of converting data into an appropriate format suitable for analysis and data mining. The objective of data transformation is to improve the usefulness and quality of data for mining algorithms. Transformation may involve:

- Scaling
- Normalization

- Encoding
- Aggregation
- Attribute Conversion

2. Difference Between Data Cleaning and Data Transformation

Cleaning	Data Transformation
Removes errors and inconsistencies	Changes the representation of data
Handles missing values	Changes scale or format
Improves data quality	Improves data usability
Performed in Experiment 3	Performed in Experiment 4

Example

Data Cleaning

Before: Attendance = ?

After: Attendance = 70

Data Transformation

Before: Attendance = 80

After: Attendance = 0.80

3. Data Normalization

Normalization is a technique used to bring attribute values into a common scale. It ensures that no attribute dominates the analysis simply because it has larger numerical values.

Example

Before Normalization:

Attribute	Value
Weight	12
Height	100
Income	50000

After Normalization:

Attribute	Normalized Value
Weight	0.10
Height	0.55

Income	0.82

4. Why Data Transformation is Required

Data transformation is performed to:

- Bring attributes into a common range.
- Improve algorithm efficiency.
- Reduce bias caused by large-value attributes.
- Improve comparison among attributes.
- Improve clustering quality.
- Improve classification performance.
- Prepare datasets for mining algorithms.

5. Benefits of Normalization

- Fair comparison of attributes.
- Improved algorithm performance.
- Reduced influence of large-value attributes.
- Better clustering and classification results.
- Improved dataset consistency.

6. Transformation Techniques Available in WEKA

- Normalize Filter
  - Converts numerical values into a normalized range.
- NumericToNominal
  - Converts numeric attributes into nominal (categorical) values.
- NominalToBinary
  - Converts nominal values into binary representation.
- Standardize Filter
  - Transforms values to have:
    - Mean = 0
    - Standard Deviation = 1

Industry Insight: In practical Machine Learning and Data Mining projects, normalized data often produces more reliable analytical results because all attributes contribute more fairly during analysis. Many machine learning algorithms such as K-Means and K-Nearest Neighbour are highly sensitive to attribute scales and therefore benefit significantly from normalization.

Mini Real-World Example:

Food Recommendation System

Consider the following dataset:

Age	Weight	Family Income
4	12	50000
5	15	60000

Without normalization: Income dominates the analysis.

Therefore, recommendation results may become biased.

After normalization: All attributes contribute proportionately.

This leads to more reliable recommendations.

Dataset Used in This Experiment:

Filename: student_transform.arff

Dataset Contents:

@relation student

@attribute attendance numeric

@attribute marks numeric

@attribute studyhours numeric

@data

80,75,5

60,40,2

90,85,6

70,60,3

50,30,1

Dataset Information:

- Relation Name: student
- Attributes: 3
- Instances: 5
- Numeric Attributes: 3
- Nominal Attributes: 0

Before Normalization vs After Normalization:

Before Normalization

Attendance	Marks	Study Hours
80	75	5
60	40	2
90	85	6
70	60	3
50	30	1

After Normalization

Attendance	Marks	Study Hours
0.75	0.82	0.80
0.25	0.18	0.20
1.00	1.00	1.00
0.50	0.55	0.40
0.00	0.00	0.00

Note: The actual normalized values may vary slightly depending on the normalization method used by WEKA.

Procedure:

Activity 1: Load Dataset

Steps:

1. Open WEKA.
2. Select Explorer.
3. Open the Preprocess tab.
4. Click Open File
5. Select student_transform.arff.
6. Click Open

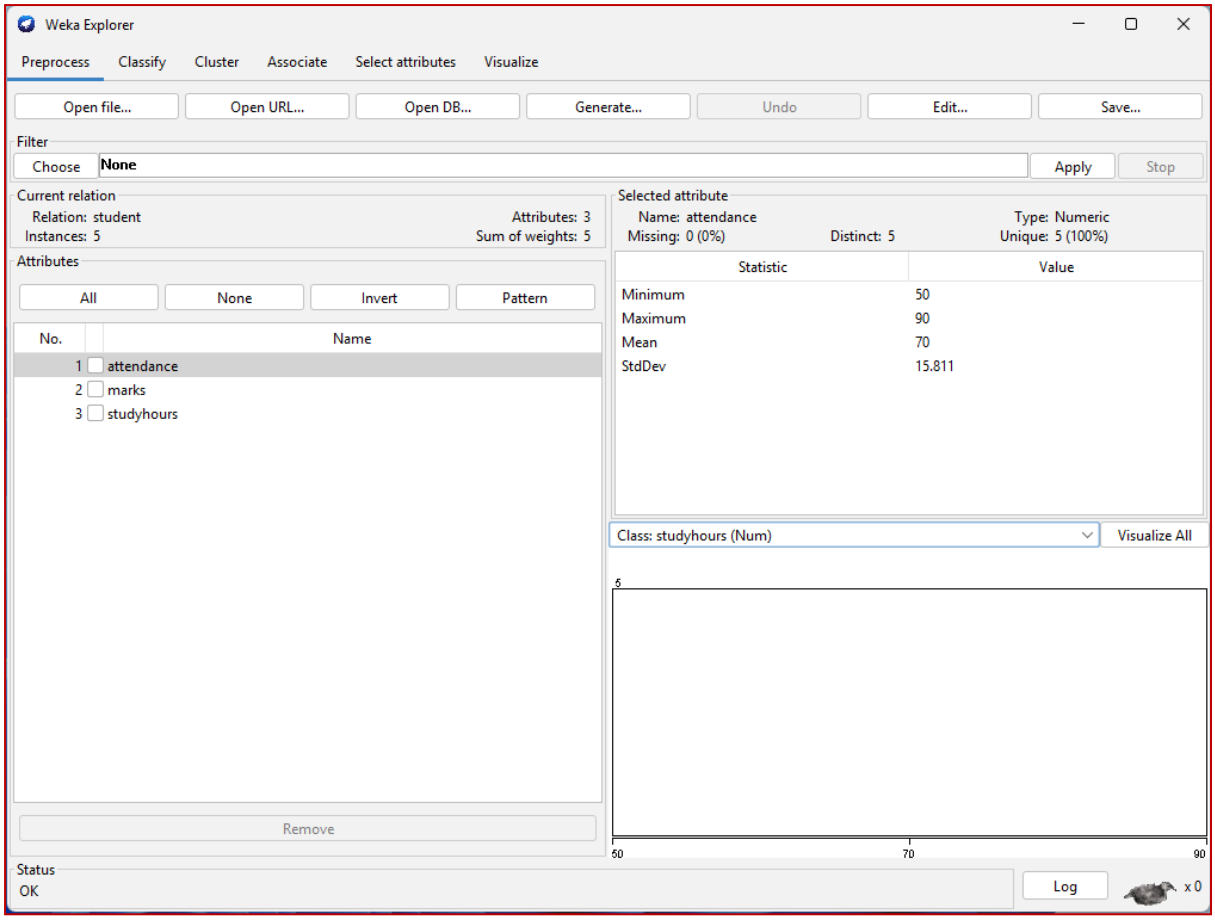


Figure 4.1: Dataset Before Transformation

Activity 2: Analyze Attribute Values

Steps:

1. Observe the values of:
   - Attendance
   - Marks
   - Study Hours
2. Record the range of values for each attribute.
3. Identify differences in scales among attributes.

Activity 3: Apply Normalize Filter

Steps:

1. Navigate to the Preprocess tab.
2. Click: Choose under the Filter section.
3. Select Unsupervised –> Attribute –> Normalize
4. Click Apply
5. Wait for WEKA to process the dataset.

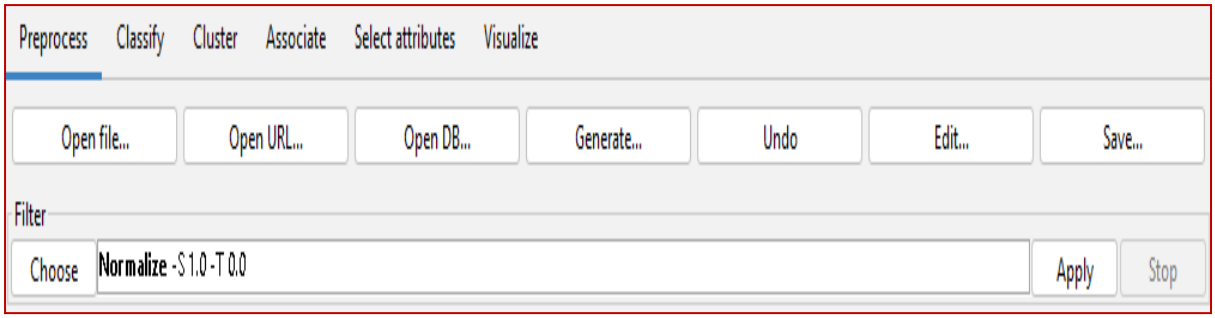


Figure 4.2: Selecting Normalize Filter

Activity 4: Observe Transformed Dataset

Steps:

1. Observe the transformed values.
2. Compare them with original values.
3. Record observations.

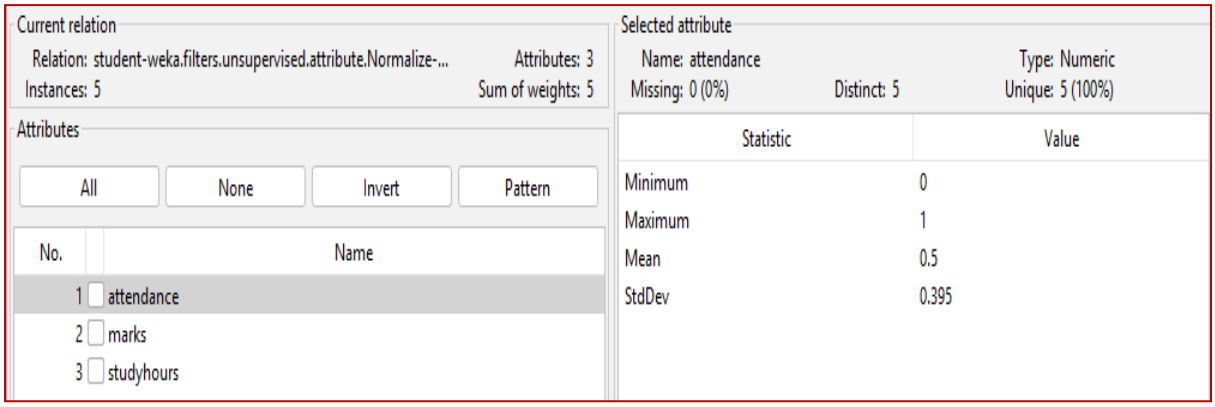


Figure 4.3: Dataset After Applying Normalize Filter

Activity 5: Compare Results

Compare:

- Original Dataset
- Normalized Dataset

Record:

- Minimum values
- Maximum values
- Difference in scales

@data	@data
80,75,5	0.75,0.818182,5
60,40,2	0.25,0.181818,2
90,85,6	1,1,6
70,60,3	0.5,0.545455,3
50,30,1	0,0,1

Figure 4.4: Comparison of Original and Normalized Values

The normalized values are scaled into a common range, making comparisons among attributes more meaningful.

Common Mistakes While Applying Normalization

1. Applying normalization repeatedly on an already normalized dataset.
   - Impact: Distorts attribute values.
2. Comparing transformed data without recording original values.
   - Impact: Transformation effects cannot be analyzed.
3. Forgetting to save the transformed dataset.
   - Impact: Work may be lost.
4. Assuming normalization changes the meaning of data.
   - Impact: Incorrect interpretation.
5. Using transformed values without understanding their significance.
   - Impact: Poor analytical conclusions.

Common Discussion

1. Why is data transformation required?
2. How is transformation different from data cleaning?
3. Why is normalization important?
4. Can normalization improve clustering performance?
5. Can large-value attributes affect mining results?
6. Which WEKA filter was used in this experiment?
7. What is the purpose of the Normalize filter?

Student Activity:

Create an ARFF dataset satisfying the following requirements:

- Minimum 5 records
- Minimum 3 numeric attributes
- Save the dataset in ARFF format.

Apply: Normalize Filter

Record:

- Original values
- Normalized values
- Minimum and maximum values before normalization
- Minimum and maximum values after normalization

Observation Table:

PARAMETER	OBSERVATION
Dataset Name	
Dataset File Name	
Number of Attributes	
Number of Instances	
Filter Applied	
Transformation Technique Used	
Attribute(s) Normalized	
Minimum Value Before Transformation	
Maximum Value Before Transformation	
Minimum Value After Transformation	
Maximum Value After Transformation	

Result: The dataset was successfully loaded and transformed using WEKA. The Normalize filter was applied to transform attribute values into a common scale. The importance of data transformation and normalization in improving data suitability for data mining applications was studied successfully.

Viva Questions:

1. What is data transformation?
2. What is normalization?
3. Why is normalization required?
4. Differentiate between data cleaning and data transformation.
5. What is the purpose of the Normalize filter?
6. What are the benefits of normalization?
7. Which WEKA tab is used for normalization?
8. What is attribute scaling?
9. Can normalization improve clustering results?
10. What is the difference between Normalize and Standardize?
11. What is NumericToNominal conversion?
12. What is NominalToBinary conversion?
13. Why are preprocessing techniques important?
14. Which attributes were normalized in this experiment?
15. How does normalization help in data mining?

Assignment Questions:

1. Explain data transformation with suitable examples.
2. Differentiate between data cleaning and normalization.
3. Explain the importance of normalization in data mining.
4. Apply normalization on a custom dataset and analyze the results.
5. Compare Normalize and Standardize filters available in WEKA.
6. Explain the effect of attribute scales on mining algorithms.
7. Create a dataset and perform normalization using WEKA.

References:

1. Ian H. Witten, Eibe Frank and Mark A. Hall, *Data Mining: Practical Machine Learning Tools and Techniques*, Morgan Kaufmann.
2. Jiawei Han, Micheline Kamber and Jian Pei, *Data Mining: Concepts and Techniques*, Morgan Kaufmann.
3. WEKA Official Documentation & User Manual.

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